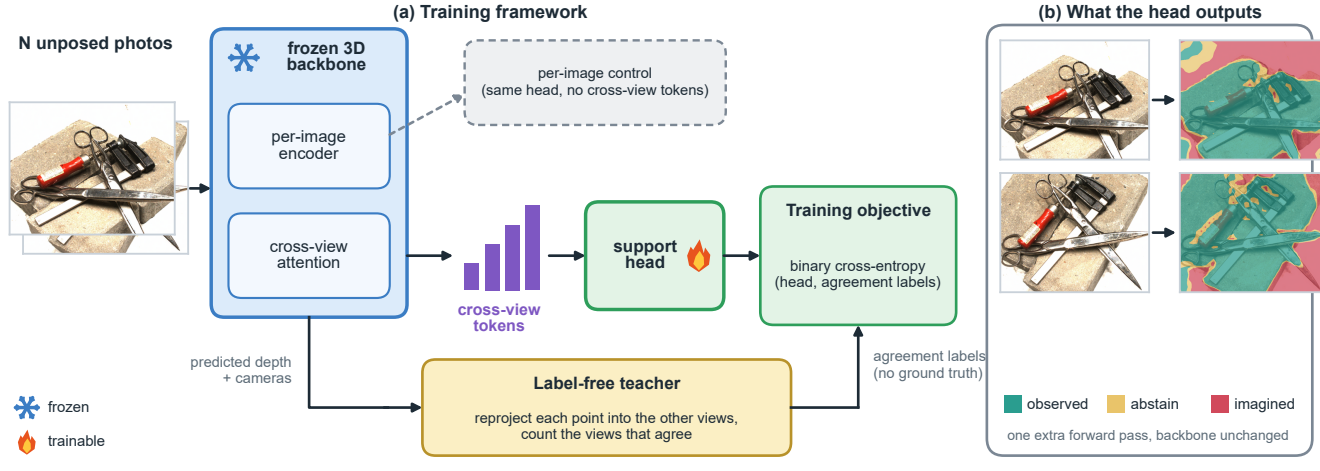

(a) Training framework
N unposed photos
frozen 3D backbone
per-image encoder
cross-view attention
per-image control
(same head, no cross-view tokens)
cross-view tokens
support head
Training objective
binary cross-entropy
(head, agreement labels)
predicted depth
+ cameras
Label-free teacher
reproject each point into the other views,
count the views that agree
agreement labels
(no ground truth)
frozen
trainable
(b) What the head outputs
observed
abstain
imagined
one extra forward pass, backbone unchanged